

Identifying customers at risk of leaving

Churn prediction

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The task

- Identify customers at elevated risk of leaving in the next six months
- Everything known on **31 December 2025**; outcome window Jan–Jun 2026
- Target avgang_6mnd is **boolean** — left, or did not
- 40,000 customers, **9.0%** of them churn — a rare positive class

- Not a churn / no-churn label — a **ranked list** by predicted probability
- Evaluated at a targeting capacity: **Precision@k** and **Recall@k**
- **Logistic regression** as an interpretable baseline
- **CatBoost** as the main model — nonlinearities and interactions in mixed numeric and categorical data, with little preprocessing

Cleaning was a substantial part of the effort:

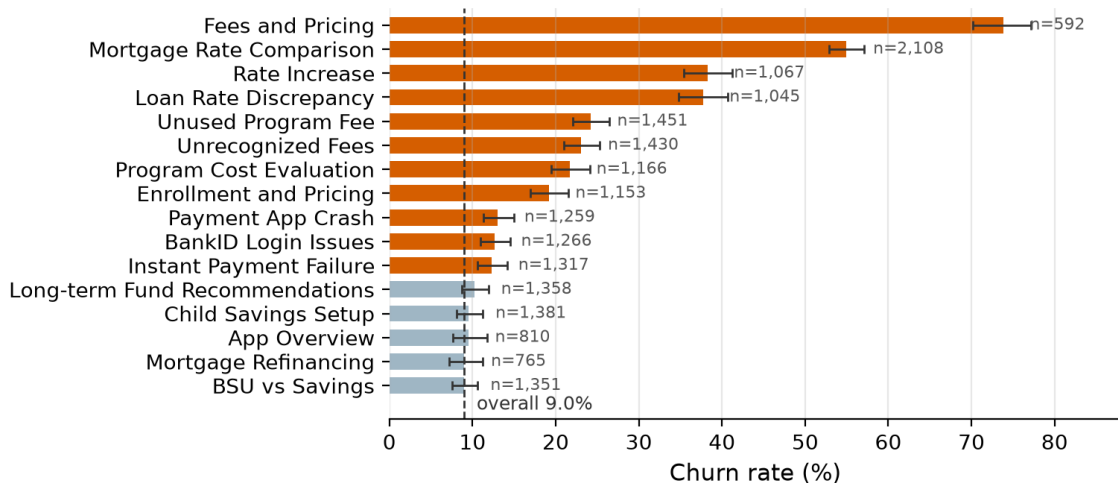
- **Duplicates** — 400 kunde_id appear twice; kept the higher advisor-call count
- **Inconsistent categories** — seven spellings of har_kundeprogram; counties in mixed case
- **Implausible values** — six ages of 121–141 set to missing; six negative deposits clipped
- **Leakage** — cancellation fees, product counts and a relative's future churn all excluded
- **Seasonality** — activity compared with the same quarter a year earlier, not with last month

- Partner and co-borrower churn is correlated
- Household pairs churn together in **1.30%** of pairs against **0.81%** expected under independence — a **1.6x** joint lift
- Prepared data for modeling: both members of a pair are held on the **same side of the train/test split**

Two text representations, compared against the same structured baseline:

- **TF-IDF** of the message text using unigrams and bigrams, with stopwords removed
- **Embedding topics** — Qwen3-Embedding-0.6B per inquiry, clustered into 16 semantic themes
- Used a locally served LLM to name each cluster from its most central messages

What the churners were asking about



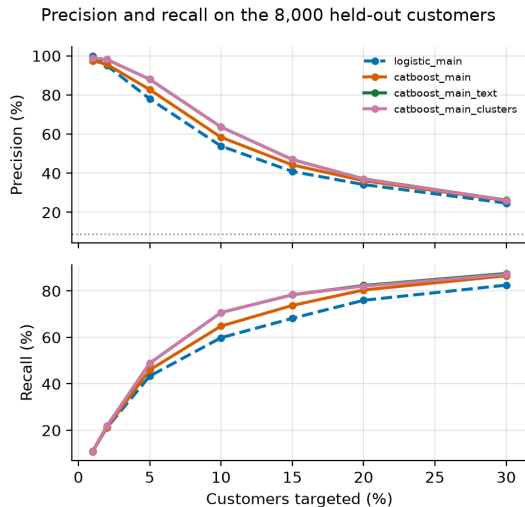
Orange where the 95% interval clears the 9.0% base rate.

Results

Model / inputs	Features	Avg. precision	Precision@10%	Recall@10%
Logistic / structured	23	0.625	53.9%	59.9%
CatBoost / structured	23	0.682	58.4%	64.9%
CatBoost / + TF-IDF	23+64	0.744	63.6%	70.7%
CatBoost / + embedding topics	23+16	0.737	63.6%	70.7%

- TF-IDF and clustering have similar performance
- Embedding topics win on interpretability

Targeting capacity



- At **5%** reach-out: **88.0%** precision, **48.9%** recall
- At **10%**: **63.6%** precision, **70.7%** recall
- Precision falls and recall rises as capacity grows
- Where to stop is an **economic** choice, not a modelling one

- The model cannot decide who to contact
- How far down the list to go has to balance the cost of a retention offer to the segment against the value of a retained customers
- The inquiry theme and engagement trend can indicate a reason and an opportunity
- We cannot estimate is how many of those customers an offer would actually retain

- Sensitive behaviour, free text and relationship data: strict **purpose limitation**, data minimization and access control
- Raw inquiry text more restricted than the derived features
- One customer's information must never be exposed to another
- **Decision support**, not an automatic basis for adverse treatment

Thank you for your time

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